NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Course Name: ETHICAL HACKING

Assignment- Week 5

TYPE OF QUESTION: MCQ/MSQ/SA

Number of questions: 10 **Total mark: 10 x 1 = 10**

QUESTION 1:

Which of the following NMAP option(s) can trigger execution of NMAP scripts?

a. -A
b. -sC
c. -Pn
d. --script=default
e. -sn

Correct Answer: a, b, d

Detailed Solution: -sC runs the default set of NSE scripts (equivalent to --script=default), and -A (aggressive scan) also enables script scanning along with OS/version detection and traceroute. -Pn only skips host discovery, and -sn only performs host discovery — neither triggers script execution.
Thus, the correct options are (a), (b), and (d).

QUESTION 2:

Which of the following tool(s)/software CANNOT be used for vulnerability scanning?

a. Nessus
b. Hydra
c. Nexpose
d. Crunch
e. NMAP

Correct Answer: b, d

Detailed Solution: Nessus, Nexpose, and NMAP (with vulnerability scripts) are used for scanning vulnerabilities. Hydra is a password-cracking/brute-force tool, and Crunch is a wordlist generator - neither performs vulnerability scanning.
Thus, the correct options are (b) and (d).

QUESTION 3:

Which of the following is NOT considered a common password cracking approach?

a. Dictionary attack
b. Brute-force attack
c. Rule-based attack
d. Packet sniffing
e. Shoulder surfing

Correct Answer: d

Detailed Solution: Dictionary, brute-force, rule-based attacks, and shoulder surfing are all recognized password cracking/acquisition approaches. Packet sniffing is a technique for capturing network traffic and is not itself a password cracking method (though captured data could later aid one).
Thus, the correct option is (d).

QUESTION 4:

Which protocol in TCP/IP is connectionless and does not provide reliability?

a. TCP
b. UDP
c. FTP
d. TELNET

Correct Answer: b

Detailed Solution: UDP sends datagrams without establishing a connection; thus, it is fast but unreliable. TCP, FTP and TELNET are connection-oriented, in which a connection is established prior to data transfer.

Thus, the correct option is (b).

QUESTION 5:

Which tool is primarily used to generate custom wordlists for use in dictionary-based password attacks?

a. Hydra
b. Nessus
c. Crunch
d. Wireshark
e. Hashcat

Correct Answer: c

Detailed Solution: Crunch is specifically used to generate custom wordlists based on rules such as character sets, length, and patterns. Hydra performs the actual brute-force/dictionary login attempts, Nessus is for vulnerability scanning, Wireshark captures packets, and Hashcat cracks password hashes rather than generating wordlists.

Thus, the correct option is (c).

QUESTION 6:

Which of the following statement(s) is/are **false** regarding malware?

a. All malware types are capable of self-replication.
b. Malware can grant partial or full control of a system to an attacker.
c. Malware can be delivered through file sharing or fake/pirated software.
d. Malware can modify, corrupt, or delete data on infected systems.
e. None of these.

Correct Answer: a

Detailed Solution: Not all malware self-replicates - for example, Trojans typically do not self-replicate, while viruses and worms do. Malware can indeed give attackers partial/full control, spread via file sharing or fake software, and modify or delete data.

Thus, the correct (false) option is (a).

QUESTION 7:

In a MAC flooding attack, what is the primary goal of overwhelming a switch's CAM table?

a. To crash the switch permanently.
b. To force the switch to behave like a hub, broadcasting traffic to all ports.
c. To increase the switch's available memory.
d. To disable the switch's power supply.
e. None of these.

Correct Answer: b

Detailed Solution: When a switch's CAM (Content Addressable Memory) table is flooded with fake MAC entries, the switch may fail to determine the correct outgoing port for legitimate traffic and start broadcasting frames out of all ports like a hub, enabling easier packet sniffing.

Thus, the correct option is (b).

QUESTION 8:

Which of the following NMAP script(s) is/are used specifically for identifying the operating system of an SMB-enabled target?

a. smb-os-discovery
b. smb-brute
c. http-os-brute
d. smb-enum-users
e. ssh-brute.

Correct Answer: a

Detailed Solution: The smb-os-discovery script is specifically designed to identify the operating system of a target via SMB. smb-brute performs credential brute-forcing, smb-enum-users performs user enumeration, ssh-brute cracks SSH credentials, and http-os-brute is not a valid/standard NMAP script.

Thus, the correct option is (a).

QUESTION 9:

An attacker is using hydra to perform a dictionary attack on an SSH service, testing a wordlist of 5000 passwords against 4 different usernames. The maximum number of logins attempts that could be made is _____.

Correct Answer: 20000

Detailed Solution: Total attempts = number of usernames x number of passwords = 4 x 5000 = 20,000 (assuming every password is tried against every username, worst case before a match is found).

Thus, the correct answer is 20000.

QUESTION 10:

If a symmetric-style brute-force password cracker needs to try all combinations of a 4-character password using only lowercase letters (26 possible characters per position), how many total password combinations are possible?

Correct Answer: 456976

Detail Solution: Total combinations = 26^4 = 26 x 26 x 26 x 26 = 456,976.
Thus, the correct answer is 456976.

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